QUBO Formulation: Credit Scorecard Combination Problem

Credit Scorecard Combination Problem

Problem Statement. Given a set of N candidate scorecards, each associated with a non-negative expected revenue contribution R_i for $i \in \{0, \ldots, N-1\}$, and a symmetric non-negative pairwise overlap penalty P_{ij} for $0 \leq i < j \leq N-1$, the objective is to select exactly K scorecards such that the total expected revenue minus the sum of pairwise overlap penalties is maximized.

Decision Variables. For each candidate scorecard $i \in \{0, \ldots, N-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$, where $x_i = 1$ indicates that scorecard i is selected, and $x_i = 0$ indicates it is not selected.

QUBO Derivation. We transform the maximization problem into a minimization problem suitable for a Quadratic Unconstrained Binary Optimization (QUBO) formulation by negating the objective and incorporating the cardinality constraint via a penalty method.

Step 1 – Original Objective. The original objective is to maximize revenue minus overlap penalties:

$$\max_{\mathbf{x}} \sum_{i=0}^{N-1} R_i x_i - \sum_{0 \leq i < j \leq N-1} P_{ij} x_i x_j. \tag{1}$$

For QUBO minimization, we equivalently minimize the negative of this expression:

$$\min_{\mathbf{x}} \left(- \sum_{i=0}^{N-1} R_i x_i + \sum_{0 \leq i < j \leq N-1} P_{ij} x_i x_j \right). \tag{2}$$

Step 2 – Constraint Formulation. The requirement to select exactly K scorecards is expressed as:

$$\sum_{i=0}^{N-1} x_i = K. \tag{3}$$

Step 3 – Penalty Term Expansion. We enforce the constraint by adding a squared penalty term scaled by a positive weight λ:

$$\text{Penalty}(\mathbf{x}) = \lambda \left(\sum_{i=0}^{N-1} x_i - K \right)^2 . \tag{4}$$

Expanding the square and applying the idempotent property $x_i^2 = x_i$ for binary variables yields:

$$\begin{aligned}
\left(\sum_{i=0}^{N-1} x_i - K \right)^2 &= \sum_{i=0}^{N-1} x_i^2 - 2K \sum_{i=0}^{N-1} x_i + K^2 + \sum_{i \neq j} x_i x_j && (5)\\
&= \sum_{i=0}^{N-1} x_i - 2K \sum_{i=0}^{N-1} x_i + K^2 + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j && (6)\\
&= (1-2K) \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j + K^2. && (7)
\end{aligned}$$

Multiplying by λ gives the general penalty contribution to the Hamiltonian:

$$\lambda \left[(1-2K) \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j + K^2 \right]. \tag{8}$$

Step 4 – Combined Hamiltonian. Adding the negated objective and the penalty term produces the full QUBO Hamiltonian $H(\mathbf{x})$:

$$H(\mathbf{x}) = \left(-\sum_{i=0}^{N-1} R_i x_i + \sum_{0 \leq i < j \leq N-1} P_{ij} x_i x_j \right) + \lambda \left[(1-2K) \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j + K^2 \right]. \tag{9}$$

Step 5 – Q Matrix Entries and Offset. Collecting linear and quadratic terms yields the standard QUBO form $H(\mathbf{x}) = \sum_{i=0}^{N-1} Q_{i,i} x_i + \sum_{0 \leq i < j \leq N-1} Q_{i,j} x_i x_j + c$, with entries defined as:

$$\begin{aligned}
Q_{i,i} &= -R_i + \lambda(1-2K), \quad \forall i \in \{0, \ldots, N-1\}, && (10)\\
Q_{i,j} &= P_{ij} + 2\lambda, \quad \forall 0 \leq i < j \leq N-1, && (11)\\
c &= \lambda K^2. && (12)
\end{aligned}$$

Penalty Weight Justification. The penalty weight λ must be chosen large enough to ensure that any violation of the cardinality constraint incurs a strictly higher energy cost than the maximum possible improvement in the objective function. The maximum objective

gain achievable by selecting $K+1$ or $K-1$ items instead of K is bounded by the sum of all individual revenues. Therefore, the condition

$$\lambda > \sum_{i=0}^{N-1} R_i \tag{13}$$

suffices to guarantee that infeasible solutions are strictly penalized relative to all feasible solutions.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^N$ recovers the optimal scorecard combination because (i) any assignment satisfying $\sum_{i=0}^{N-1} x_i = K$ incurs zero penalty, reducing $H(\mathbf{x})$ to the negated original objective; (ii) any infeasible assignment incurs a penalty of at least λ, which by construction exceeds the maximum possible reduction in the objective term achievable by any feasible assignment. Consequently, the global minimum of $H(\mathbf{x})$ must correspond to a feasible solution that minimizes the negated objective, which is equivalent to maximizing the original revenue-minus-penalties objective.

Illustrative Example. Consider a concrete instance with $N = 3$ candidate scorecards and $K = 2$ required selections. Let the revenues be $R_0 = 10$, $R_1 = 20$, and $R_2 = 30$. Let the pairwise overlap penalties be $P_{01} = 5$, $P_{02} = 8$, and $P_{12} = 12$. The sum of revenues is $10 + 20 + 30 = 60$, so we choose $\lambda = 61$ to satisfy the penalty weight condition.

Substituting these values into the general QUBO parameters yields:

$$Q_{0,0} = -10 + 61(1-4) = -193, \tag{14}$$
$$Q_{1,1} = -20 + 61(1-4) = -203, \tag{15}$$
$$Q_{2,2} = -30 + 61(1-4) = -213, \tag{16}$$
$$Q_{0,1} = 5 + 2(61) = 127, \tag{17}$$
$$Q_{0,2} = 8 + 2(61) = 130, \tag{18}$$
$$Q_{1,2} = 12 + 2(61) = 134, \tag{19}$$
$$c = 61 \times 2^2 = 244. \tag{20}$$

The resulting Hamiltonian is:

$$H(x_0, x_1, x_2) = -193x_0 - 203x_1 - 213x_2 + 127x_0x_1 + 130x_0x_2 + 134x_1x_2 + 244. \tag{21}$$

Enumerating the three feasible assignments ($\sum x_i = 2$):

$$H(1,1,0) = -193 - 203 + 127 + 244 = -25, \tag{22}$$
$$H(1,0,1) = -193 - 213 + 130 + 244 = -32, \tag{23}$$
$$H(0,1,1) = -203 - 213 + 134 + 244 = -38. \tag{24}$$

The global minimum is -38 at $\mathbf{x}^* = (0,1,1)$. This corresponds to selecting scorecards 1 and 2, yielding a maximum total score of $R_1 + R_2 - P_{12} = 20 + 30 - 12 = 38$, which matches the negated minimum energy $-(-38) = 38$.